

Why Flow Matching Fails at Missing-Modality Brain Tumour Segmentation: A Diagnostic Study

Author One*, Author Two*, Author Three†

*Affiliation One, City, Country

Email: {author.one, author.two}@example.edu

†Affiliation Two, City, Country

Email: author.three@example.edu

Abstract—Generative segmentation models are attractive for missing-modality medical imaging because they can be conditioned on whatever subset of modalities is available. We test that promise on 3D brain tumour segmentation, where every training sample has exactly one of four MRI modalities withheld and the target is the full four-class tumour label map. Holding the backbone, data pipeline and optimiser fixed, we compare a discriminative 3D U-Net (A) with a flow-matching model (B). On a binary whole-tumour target B stays within 0.10 Dice of A (0.779 vs. 0.878) but costs $\approx 14\times$ more to sample. On the four-class target it collapses to 0.553 whole-tumour Dice against A’s 0.824, and all 36 patient-paired comparisons over three seeds, four missing patterns and three regions are negative with bootstrap intervals excluding zero. We then report a pre-registered four-part diagnostic that rules out overfitting, capacity, numerical integration and background-distribution mismatch, and localises the failure to two mechanisms: the regression target $v^* = (\mathbb{E}[y_1 | y_t] - y_t)/(1 - t)$ amplifies posterior error without bound as $t \rightarrow 1$, and the voxel-wise L_2 objective is dominated by the $\sim 84\%$ background voxels, so the learned field is accurate in aggregate while the region that decides the argmax is six to eight times worse. Increasing the number of integration steps makes the result worse, not better. We then test loss reweighting along both diagnosed axes. PLACEHOLDER — remedy results pending. We conclude that for near-deterministic, strongly class-imbalanced 3D targets the failure is a property of the objective and its parameterisation rather than of data volume or compute budget, and that the cost–accuracy trade-off of flow matching is unfavourable in this regime.

Index Terms—flow matching, missing modalities, brain tumour segmentation, generative segmentation, failure analysis, BraTS

I. INTRODUCTION

Missing modalities are the normal case in clinical MRI, not the exception. A patient may arrive without contrast-enhanced T1 because of renal function, or without FLAIR because of acquisition time; the segmentation model still has to produce a label map. A large literature therefore trains segmentation networks that accept an arbitrary subset of modalities, typically by imputing the absent volumes or by learning a shared latent representation [1]–[5].

Generative models have recently been proposed for this setting on the grounds that a conditional generative model trained on the full-modality distribution can be evaluated under any conditioning subset, and that it models the segmentation as a distribution rather than a point prediction. Flow matching (FM) [6], [7] is an appealing instance of this idea: it is simulation-free to train, has a simple regression objective, and requires no noise schedule tuning beyond a choice of interpolation path. FM and diffusion variants have been reported to work well on medical segmentation benchmarks [8]–[11].

This paper is a diagnostic study rather than a new architecture. We take a concrete missing-modality task — four-class brain tumour segmentation from BraTS when exactly one of T1, T2, T1ce and FLAIR is withheld — and we hold everything except the model class fixed: same 3D U-Net backbone, same parameter count to within 0.16%, same patients, same patches, same augmentations, same missing-modality schedule, same optimiser budget, and a data pipeline whose randomness is a pure function of (seed, step) so that both models see identical samples.

We find a sharp split. On the *binary* whole-tumour target the FM model stays within 0.10 Dice of a discriminative model of the same capacity. On the *four-class* target it collapses, losing 0.27 whole-tumour Dice and up to 0.37 enhancing-tumour Dice, with every one of the 36 paired comparisons over three seeds negative at a 95% bootstrap interval that excludes zero. Because the two tasks differ only in the number of output classes, the collapse is not explained by data, capacity or budget.

We then run a pre-registered four-part diagnostic to localise the cause, and we report what it excludes as carefully as what it implicates. Overfitting, capacity, numerical integration error and background-distribution mismatch are each ruled out by an independent measurement. The surviving explanation is a pair of mechanisms that are visible only when the target is near-deterministic and strongly class-imbalanced: the flow’s regression target is unboundedly sensitive to posterior error near $t = 1$, and the L_2 objective is dominated by background voxels, so the model is accurate on average while its errors

concentrate where the argmax is decided rather than in near-ties.

PLACEHOLDER — remedy intro pending.

Our contributions are:

- A controlled comparison, with an identical backbone and an identical data stream, showing that flow matching is competitive on a binary tumour target but collapses on a four-class target under missing modalities (Section V).
- A pre-registered diagnostic that excludes overfitting, capacity, integration error and background mismatch as explanations, and identifies the $1/(1-t)$ error amplification and background-dominated L_2 as the mechanisms (Section V-C).
- A test of two loss reweightings derived from those mechanisms, evaluated at three seeds against the same baseline.
- A cost and reproducibility accounting, including a measured patient-wise inference cost of 6.06 s versus 0.42 s and a demonstration that repeated runs of the FM model diverge chaotically from a bit-identical start (Section VI).

II. RELATED WORK

a) Missing-modality segmentation.: [1] introduced HeMIS, which aggregates statistics of available modalities so that any subset can be handled at test time. [2] learn a shared latent space with a hetero-modal variational autoencoder and reconstruct absent modalities. [3] combine a transformer with modality-specific encoders to cope with incomplete multi-modal input. [12] survey the area. These methods are discriminative: they predict a label map directly and are trained with a Dice/CE objective [13]. Our model A is an instance of this family and serves as the reference point; our subject is instead a generative alternative trained with a flow-matching objective.

b) Flow matching and diffusion for segmentation.: Denoising diffusion [14] and score-based generative models [15] generate samples by integrating a learned field; flow matching [6] and rectified flow [7] replace the stochastic process by an ordinary differential equation with a simple conditional regression target. [8] and [9] apply diffusion to image segmentation, and report gains on 2D benchmarks. Practical parameterisations and loss weightings are studied in [16] for diffusion and [17] for rectified flow; the distinction between predicting the velocity and predicting the clean target is known to matter at high noise levels [18]. This paper quantifies that trade-off’s consequences in a near-deterministic, strongly class-imbalanced 3D setting, where it is decisive.

c) Imbalance and evaluation.: Dice [19], focal [20] and Tversky [21] losses exist precisely because voxel-wise objectives are dominated by background. Our diagnostic isolates the analogous effect for a generative objective, where it is compounded by the amplification described in Section III-C because the regression target and the decision rule live in different spaces. Calibration and overconfidence are studied in [22]; the endpoint margins we measure in Section V-C are the generative analogue.

III. METHOD

A. Task and data

We use the BraTS 2020 training set [23]–[25], of which we take a fixed subset of 100 patients and split them 60/20/20 by patient with a fixed seed. All volumes are $240 \times 240 \times 155$ at 1 mm isotropic and are already co-registered; we reorient to RAS and do *no* resampling and *no* cropping. Retaining the full field of view is deliberate: any crop would be a function of either the ground-truth tumour mask or of the modality that is later withheld, which would leak information into the missing-modality setting. Each modality is z -scored inside its own non-zero brain support and set to zero outside it, using per-volume statistics only.

The missing modality is simulated *after* normalisation by zeroing the corresponding channel, and a four-channel availability indicator is concatenated so that the model can tell “absent” from “present and zero”. During training each sample hides exactly one of {T1, T2, T1ce, FLAIR} with equal probability; at test time every patient is evaluated under all four missing patterns. The target has four mutually exclusive classes (background, NCR/NET, ED, ET). We report Dice for the derived regions $WT = \{1, 2, 4\}$, $TC = \{1, 4\}$ and $ET = \{4\}$ of the original label values.

B. Two models, one backbone

Both models use the same 3D U-Net [26]: four levels with [16, 32, 64, 128] channels, two $3 \times 3 \times 3$ convolutions per block, GroupNorm, SiLU, max-pooling downsampling and transposed-convolution upsampling with skip connections. Model A takes 4 MRI channels plus 4 availability channels and outputs K class logits ($K = 4$), trained with the mean soft Dice over the non-background classes plus a four-class cross-entropy, equally weighted. Model B takes the same 8 channels plus K noisy-mask channels and one constant time channel, and outputs a K -channel velocity field trained with a pure mean-squared-error objective. Parameter counts are 1,404,340 (A) and 1,406,500 (B); the difference of 2,160 is exactly the five extra input channels in the first convolution, so the representational capacity is matched to within 0.16%.

C. Flow matching and the two mechanisms

Let y_1 be the one-hot label map and $y_0 \sim \mathcal{N}(0, I)$ the same shape. We use the linear interpolant

$$y_t = (1-t)y_0 + ty_1, \quad t \in [0, 1], \quad (1)$$

and train the network to regress the conditional velocity $v = y_1 - y_0$ given (y_t, t) and the MRI conditioning. At convergence the learned field is the conditional expectation

$$v^*(y_t, t) = \mathbb{E}[y_1 | y_t] - \mathbb{E}[y_0 | y_t] = \frac{\mathbb{E}[y_1 | y_t] - y_t}{1-t}. \quad (2)$$

Sampling integrates $dy/dt = v_\theta(y, t)$ from $t = 0$ to $t = 1$ with 16 Euler steps and takes the argmax only after the last update.

Equation (2) exposes both mechanisms we investigate. First, the sensitivity of the regression target to the posterior estimate is

$$\frac{\partial v^*}{\partial \mathbb{E}[y_1 | y_t]} = \frac{1}{1-t}, \quad (3)$$

which diverges as $t \rightarrow 1$: an error ε in the posterior produces a velocity error $\varepsilon/(1-t)$. The last steps of the ODE are therefore the steps that convert small model errors into trajectory error. Second, the objective is a voxel-wise L_2 and therefore weights every voxel equally, while background occupies roughly 84% of the sampled volume (Section V-C). A field can be excellent in aggregate and still decide the argmax wrongly.

D. Remedies

The two mechanisms suggest two orthogonal reweightings of the same L_2 objective. We weight each voxel by

$$w(v, t) = (1 + \alpha \mathbb{I}[y_1(v) \neq 0]) (1 - t + \epsilon)^{-\gamma}, \quad (4)$$

where the first factor raises the weight of foreground voxels by $1 + \alpha$ (addressing the background domination) and the second emphasises late times, where the amplification of Eq. (3) acts. We use $\alpha = 9$, $\gamma = 1$ and $\epsilon = 0.1$, which bounds the temporal factor by 10. The weighted loss is normalised by the sum of weights so that its scale is comparable to the unweighted mean. We evaluate the foreground term, the temporal term and their combination, each at three training seeds. Setting $\alpha = \gamma = 0$ reproduces the baseline exactly.

IV. EXPERIMENTAL SETUP

a) Training.: 10,000 optimiser updates per run, batch size 4, patches of 64^3 drawn either around the tumour (50%) or uniformly from the brain support, AdamW with learning rate 10^{-4} and weight decay 10^{-5} , gradient norm clipped at 12. Augmentation is random flipping plus per-channel scaling and shifts plus Gaussian noise; noise is applied only to available modalities so that withheld channels stay exactly zero. Data and model noise are drawn from separate generators that depend only on (*seed*, *step*), so A and B see identical patients, patches, augmentations and missing patterns. Three training seeds (0, 1, 2) per configuration.

b) Validation and checkpoint selection.: Validation runs every 2,000 updates with each validation patient assigned a single, fixed missing pattern, balanced across patterns and independent of the training seed. The selection score is the mean over the four missing patterns and three regions of the per-patient mean Dice. After training, the selected checkpoint is re-evaluated on all validation patients and all four patterns without re-selection.

c) Metrics and statistics.: We report Dice and recall, and the 95th percentile of the symmetric surface distance (HD95) at 1 mm spacing [27]. When both prediction and reference are empty we set Dice to 1; when exactly one is empty we set Dice to 0 and exclude the case from the HD95 mean, reporting the number of such failures. For every (seed, missing pattern, region) cell we compute patient-paired differences B – A and a bootstrap interval over 2,000 patient resamples. The bootstrap

covers evaluation noise only, not training-seed variance, and we make no multiple-comparison correction across the 36 cells.

d) Pre-registered readouts.: Before running the diagnostic we fixed five decision rules: (i) a small-sample fit below 0.65 Dice indicates the field-and-sampling path cannot represent the task; (ii) a train-test gap above 0.10 indicates a generalisation problem; (iii) a change below 0.05 when the MRI conditioning is shuffled indicates the model is not truly conditioned; (iv) a change below 0.02 between 16 and 64 Euler steps indicates integration is not the bottleneck; (v) a still-rising validation curve at 40,000 steps indicates undertraining.

V. RESULTS

A. Binary target: competitive but expensive

Table II shows the binary whole-tumour task. Flow matching is 0.09–0.10 Dice behind the discriminative model across all three input scenarios, including the complete-modality scenario, and the gap does not grow as modalities are removed. It is however 14 times slower to sample (5.93 s vs. 0.39 s per volume). This is the regime in which the generative approach is usually reported: a single foreground/background decision, where the target is far from deterministic at the voxel level and the decision region is a smooth surface.

B. Four-class target: collapse

Table I shows the four-class task. Model B reaches 0.553 whole-tumour Dice against A’s 0.824, 0.360 against 0.698 for the tumour core, and 0.228 against 0.601 for the enhancing tumour. All 36 patient-paired comparisons across three seeds, four missing patterns and three regions are negative, and every bootstrap interval excludes zero; the smallest gap is -0.156 Dice and the largest is -0.478 . The degradation is not confined to a particular missing modality, although removing FLAIR is worst for B (0.343 vs. 0.741 whole-tumour Dice, against 0.621–0.627 for the other three patterns), consistent with FLAIR carrying the most independent information for this target.

The failure is also qualitatively different from a loss of accuracy. Model B fills the volume with hundreds of small false-positive components: the median number of connected components in its whole-tumour mask is 332–2506, against 4–50 for model A, and 5.7–45.9% of its false positives lie outside the brain, against 0.2–3.4% for A. Figure 1 shows the shape of the error on one patient: B inflates the necrotic core far beyond the reference and scatters false-positive islands well outside the brain, while A tracks the reference closely. This is not uncertainty around a correct segmentation; it is a systematically wrong one.

C. Diagnostic: what the collapse is not

a) Check 1 — not overfitting, not capacity.: Fine-tuning on two training patients with balanced patch sampling, model A reaches 0.941 whole-tumour Dice on those patients with a median of 2 connected components, while model B reaches

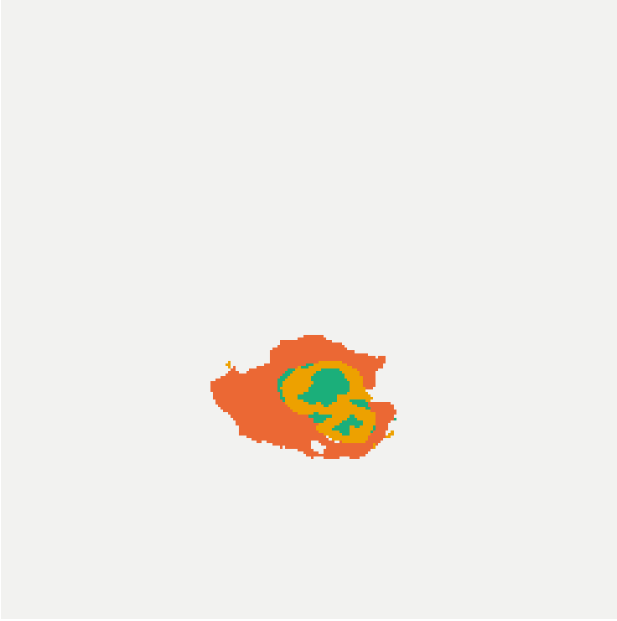
TABLE I

FOUR-CLASS MISSING-MODALITY SEGMENTATION ON THE BRA5 TEST SPLIT (20 PATIENTS, MEAN $\pm$ STD OVER THREE TRAINING SEEDS). DICE AND RECALL ARE COMPUTED PER PATIENT THEN AVERAGED; HD95 IS IN MM. “BLOBS” IS THE MEDIAN NUMBER OF CONNECTED COMPONENTS OF THE PREDICTED WHOLE-TUMOUR MASK.

Missing	Region	Dice $\uparrow$		HD95 $\downarrow$		Sec./vol. $\downarrow$	
		A (disc.)	B (FM)	A	B	A	B
w/o T1	WT	0.850 ± 0.011	0.627 ± 0.064	12.9	72.8	0.42	6.06
w/o T1	TC	0.739 ± 0.032	0.415 ± 0.055	13.7	63.1	0.42	6.06
w/o T1	ET	0.664 ± 0.030	0.284 ± 0.044	9.2	51.1	0.42	6.06
w/o T2	WT	0.847 ± 0.009	0.621 ± 0.023	15.8	90.2	0.41	6.05
w/o T2	TC	0.747 ± 0.014	0.391 ± 0.046	18.7	69.6	0.41	6.05
w/o T2	ET	0.690 ± 0.033	0.288 ± 0.054	10.3	56.3	0.41	6.05
w/o T1ce	WT	0.859 ± 0.018	0.622 ± 0.063	15.2	76.4	0.41	6.04
w/o T1ce	TC	0.631 ± 0.014	0.350 ± 0.020	20.3	57.9	0.41	6.04
w/o T1ce	ET	0.422 ± 0.038	0.122 ± 0.044	15.5	52.7	0.41	6.04
w/o FLAIR	WT	0.741 ± 0.053	0.343 ± 0.075	22.6	92.8	0.41	6.05
w/o FLAIR	TC	0.675 ± 0.048	0.285 ± 0.087	26.9	76.3	0.41	6.05
w/o FLAIR	ET	0.631 ± 0.007	0.217 ± 0.061	10.0	70.9	0.41	6.05

BraTS20_Training_027, missing flair, axial slice 68 (NCR/NET, ED, ET)

A: discriminative



B: flow matching

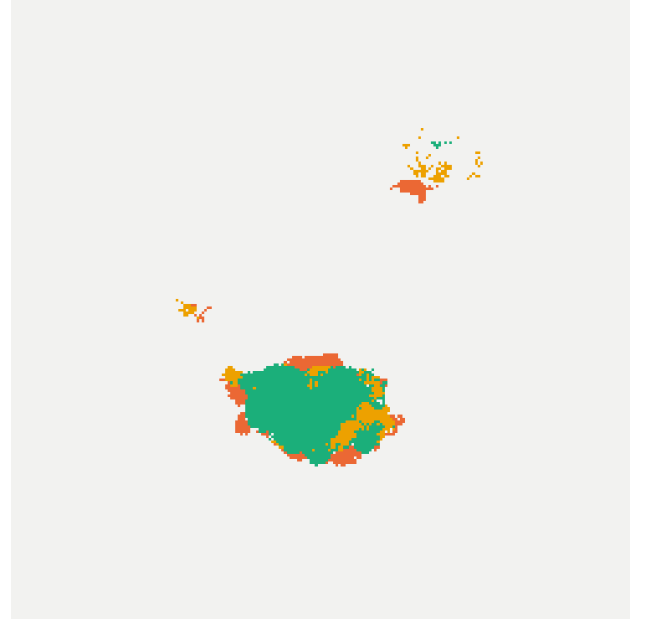


Fig. 1. One test patient with FLAIR withheld, axial slice through the tumour. Model B over-segments the necrotic core and places false-positive islands far from the tumour, while A follows the reference. Colours encode the three foreground classes; the background is unfilled.

0.538 with 5,569 components. On the full test split the train-to-test gap for B is only 0.058, below the pre-registered 0.10 threshold. The task is fittable — A demonstrates it — and B’s deficit is present on the training patients themselves, so it is not a generalisation problem.

b) Check 2 — the model is conditioned, and the field is accurate on average.: Replacing the MRI conditioning with zeros or with a shuffled volume drives B’s Dice to 0.000, and substituting another patient’s MRI leaves it at 0.049, against 0.553 with the real conditioning: B is genuinely using the

MRI, well beyond the pre-registered 0.05 margin. On held-out patches the velocity field has R^2 between 0.855 and 0.966 against the conditional target, and beats a global-class-prior reference by factors of 1.8 to 6 in the low and middle range of t . In L_2 terms the field is good.

c) Check 3 — the error is not uniform, and it is not the background distribution.: The velocity error is not spread evenly over t : per-step RMS error rises by a factor of 1.8 from 0.231 at $t \approx 0.06$ to 0.416 at $t \approx 0.94$, while the target variance is constant at 1.188. Equivalently, R^2 falls

TABLE II
BINARY WHOLE-TUMOUR TASK (THREE SEEDS, 20 TEST PATIENTS). FLOW
MATCHING STAYS WITHIN ~ 0.10 DICE HERE, UNLIKE THE FOUR-CLASS
TASK IN TABLE I.

Scenario	A Dice	B Dice	A sec.	B sec.
C0 (all)	0.878 ± 0.005	0.779 ± 0.010	0.39	5.93
C1 (w/o T1ce)	0.874 ± 0.006	0.780 ± 0.014	0.38	5.93
C2 (T2+FLAIR)	0.870 ± 0.008	0.776 ± 0.015	0.37	5.93

from 0.955 to 0.855 as $t \rightarrow 1$ — exactly the direction predicted by the $1/(1-t)$ factor of Eq. (3). The error is also not uniform across classes: at $t = 0.562$ the per-class MSE is 0.024 for background against 0.164, 0.143 and 0.191 for NCR, ED and ET. Aggregate MSE therefore understates the error in the region that decides the argmax by roughly six to eight times. Both effects are visible in Figure 2. We tested the alternative explanation that B hallucinates because its background distribution is mismatched: an augmentation variant that keeps the background strictly zero raises B’s test whole-tumour Dice by 0.007 (+0.022 for A), so the pre-registered 0.05 threshold is not met.

d) *Check 4 — integration is not the bottleneck, and more steps are worse.*: Sweeping Euler steps from 8 to 128 moves whole-tumour Dice monotonically *down* from 0.560 to 0.483; the pre-registered threshold for “integration is not the bottleneck” was a change below 0.02, and we observe -0.034 in the direction opposite to the one that a solver-error explanation would require. A related measurement explains why: the median margin between the largest and second-largest channel of the final continuous state is 0.984 and is invariant to step count (0.983 at 8 steps and 0.984 at 128), brain and non-brain alike (0.978 vs. 0.984), and only 1.2% of voxels have a margin below 0.25. Voxels that turn out to be wrong do carry a lower mean margin (0.52 against 0.97 for correct ones), so the margin is weakly informative; but because almost no voxel is near a tie, the errors cannot be dismissed as unresolved ambiguity. The endpoint is not a coin flip that better integration would settle — a more accurate solve of the same field simply follows the trajectory further into the wrong answer. Figure 3 shows both effects.

e) *Check 5 — undertraining explains only a quarter of the gap.*: Training B for 40,000 steps instead of 10,000 raises test whole-tumour Dice from 0.529 to 0.599, closing about a quarter of the gap to A, while A reaches 0.803 with a quarter of the updates. Model B’s 40k-step curve is also non-monotonic, with a collapse at 30,000 steps.

D. Remedies

PLACEHOLDER — remedy results pending.

VI. DISCUSSION

a) *Cost.*: Sampling B costs 6.06s per volume against 0.42s for A, a factor of 14.4, and training costs 28.1 min against 11.1 min. Peak GPU memory is 8.79 GB against 7.19 GB. Both models fit in a single consumer GPU, but the

generative model gives up 0.27, 0.34 and 0.37 Dice on WT, TC and ET in exchange for $14\times$ the inference latency and $2.5\times$ the training time. In a clinical pipeline where a volume must be segmented per patient per session, this is the dominant practical consideration.

b) *Reproducibility.*: Re-running the same configuration with only the validation interval changed reproduces the first 100 updates bit-for-bit — the data and noise generators are deterministic functions of (seed, step) — but from update 200 a 10^{-7} floating-point difference is amplified to 1–6% and saturates. Two runs of the identical configuration therefore differ by about ± 0.02 absolute Dice in validation score. A 40,000-step run is consequently an independent run rather than an extension of a 10,000-step one, and part of the spread across seeds reflects this chaos rather than a property of the model. We report it because single-seed comparisons of generative segmentation models at this scale are not reliable.

c) *Limitations.*: The study uses 100 patients from a single dataset and a single backbone; the four-class conclusion may not transfer to larger models or to datasets whose targets are less class-imbalanced. The endpoint-margin measurement covers three patients and two missing patterns for one seed. We diagnose and test remedies but do not claim a working fix unless the remedy results above support it. Our HD95 follows the implementation in our codebase — the 95th percentile of the pooled symmetric surface distances — and is not the official BraTS scoring implementation, so HD95 values should be read comparatively rather than as absolute benchmark numbers. Finally, the bootstrap covers evaluation noise only; training-seed variance is reported separately and the 36 cells are not corrected for multiple comparisons.

VII. CONCLUSION

We set out to test whether flow matching is a useful generative alternative for missing-modality 3D brain tumour segmentation, and found a sharp regime boundary. On a binary whole-tumour target it is competitive at $15\times$ the sampling cost; on a four-class target it fails by up to 0.37 Dice, uniformly across seeds, missing patterns and regions. A pre-registered diagnostic rules out overfitting, capacity, integration error and background mismatch, and locates the cause in the interaction between an unboundedly sensitive regression target and a background-dominated L_2 objective. The practical implication for practitioners is that a flow-matching objective should not be adopted for near-deterministic, strongly class-imbalanced volumetric targets without checking both mechanisms, and that inference cost should be counted as part of the comparison.

PLACEHOLDER — remedy conclusion pending.

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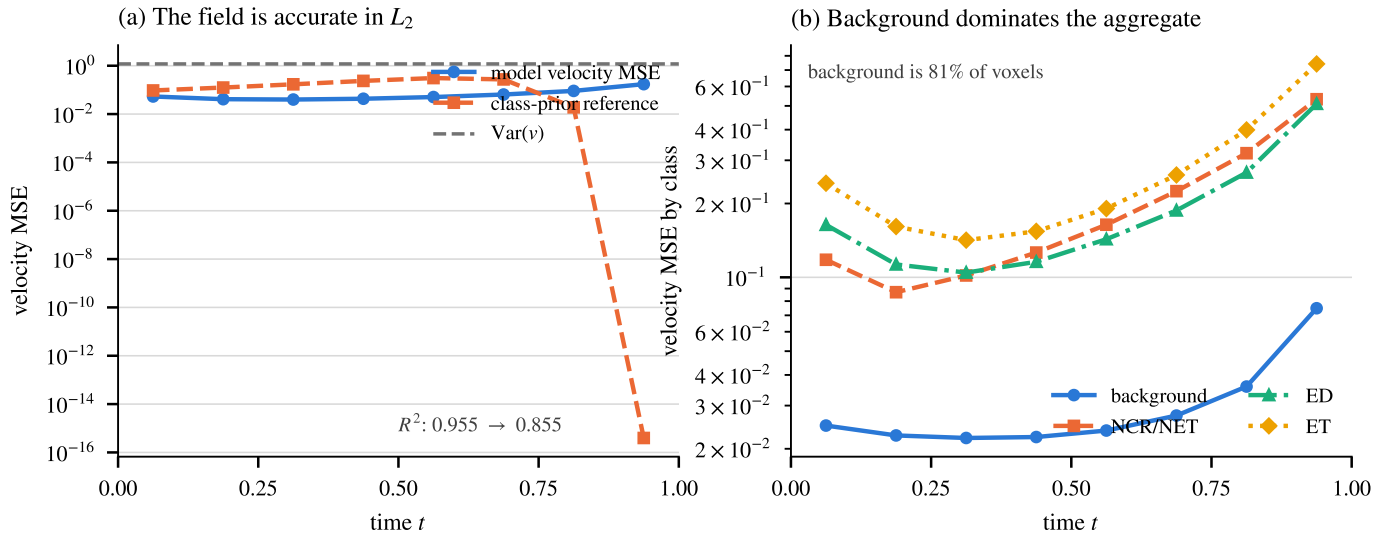
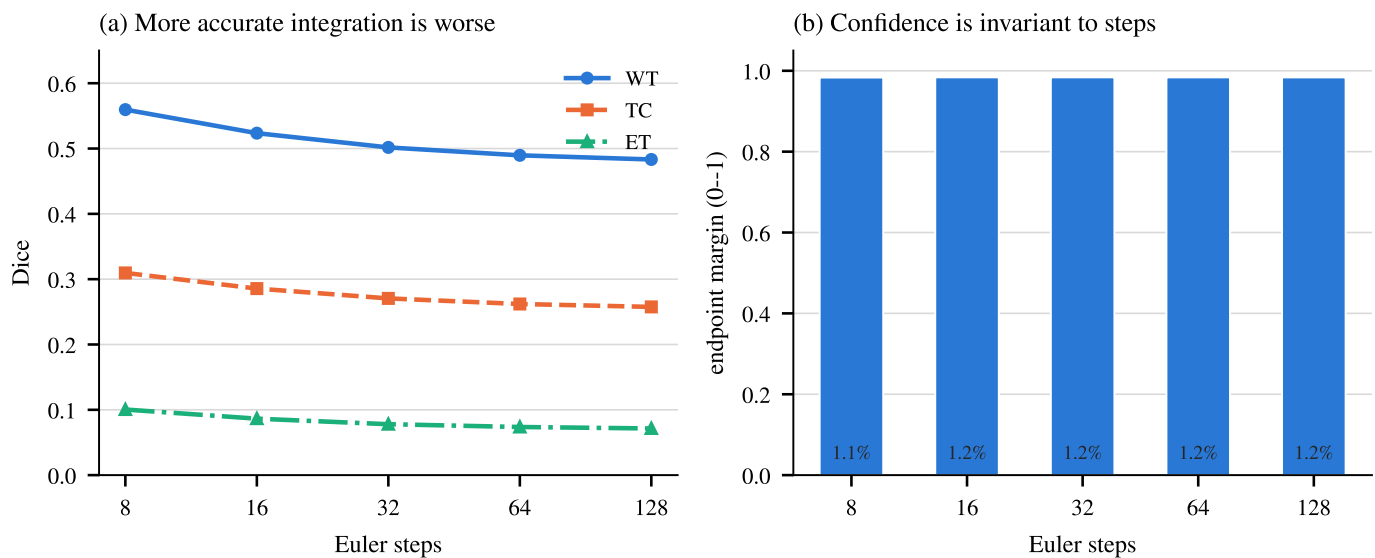


Fig. 2. (a) The learned velocity field is accurate in L_2 (note the log scale): its MSE is far below the target variance and, in the low and middle range of t , below a class-prior-only reference. The reference is meaningless once y_t nearly determines the answer, which is why that curve is not drawn past $t \approx 0.8$. (b) The same error split by class: background sits two orders of magnitude below the tumour classes, so the aggregate MSE in (a) understates the error in the region that decides the argmax.

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values below bars: share of voxels with margin < 0.25

Fig. 3. (a) Whole-tumour, tumour-core and enhancing-tumour Dice as a function of the number of Euler steps: more accurate integration of the same field is monotonically worse. (b) Mean endpoint decision margin, split by whether the voxel is ultimately classified correctly: both curves are flat across step counts. Wrong voxels are less confident than correct ones, so the margin carries some information, but almost no voxel is near a tie.

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